Algebra 1

OPTIMA ACADEMY ONLINE · FAMILY COURSE OVERVIEW · FLORIDA COURSE 1200310 · HONORS 1200320 · FULL YEAR

~4¼ hrs per week
50 min/day, M–F

About this overview. This document is an example of how the course might be taught. Scholars may see some variation in the lessons, examples, and practice sets from course to course, and between the live school and On-Demand. Each teacher also leans into their own strengths and passions; the structure, expectations, and time commitment below are representative of the course as designed.

Algebra 1 is the foundation of high school mathematics. Scholars learn to model the world with the three great function families: linear, quadratic, and exponential. The first semester builds fluency with exponents and radicals, equations and inequalities, linear functions, and systems; the second semester opens up polynomials and factoring, quadratics, exponential growth and decay, and the analysis and comparison of functions, closing with a unit on real-world data and statistics.

THE YEAR AT A GLANCE

- **M1** **Laws of Exponents & Radicals** (~2½ wks) — rational exponents; operations on radicals
- **M2** **One-Variable Equations & Inequalities** (~3½ wks) — equations, compound inequalities, absolute value
- **M3** **Linear Equations & Scatter Plots** (~4 wks) — graphing, writing equations, lines of fit
- **M4** **Linear Inequalities & Absolute Value** (~2 wks) — two-variable graphs and equations
- **M5** **Systems of Equations & Inequalities** (~1½ wks) — graphing, substitution, elimination, constraints
- **M6** **Algebraic Reasoning** (~3 wks) — polynomial operations and factoring
- **M7** **Quadratic Equations & Functions** (~2½ wks) — graphing, writing, and solving quadratics
- **M8** **Exponential Functions** (~2½ wks) — growth, decay, compound interest
- **M9** **Analyzing Functions** (~2 wks) — classifying, evaluating, rate of change
- **M10** **Comparing & Transforming Functions** (~2 wks) — linear vs nonlinear; transformations
- **M11** **Statistical Data** (~2 wks) — correlation, surveys, two-way tables

Honors (1200320). Honors scholars complete additional lessons woven through the course, including absolute value inequalities, residual analysis for lines of fit, combining functions, advanced transformations, and relative frequency tables.

A TYPICAL WEEK

- A new lesson most days: teacher-guided instruction with worked examples and notes (about 30–35 min).
- Practice problems with every lesson, worked out on paper, to make each skill automatic (about 15–20 min).
- A review and module assessment as each module closes, roughly every two to three weeks.

TIME COMMITMENT

250 minutes / week

≈ 50 minutes a day, Monday–Friday

- Daily lesson + practice fills most of the 50 minutes.
- Module review and assessment days replace new lessons as each module closes.

WHAT SCHOLARS NEED

- A computer with reliable internet and access to Canvas.
- A dedicated math notebook, pencil, and paper — every problem gets worked by hand.
- A scientific calculator; graphing happens often, on paper and with digital tools.
- Microsoft 365 and OneDrive (provided to every scholar) for any typed work.

On-Demand (asynchronous). Scholars move through the course at their own pace, with due dates to keep the year on track. An Optima teacher is available throughout for support, feedback, and grading.

Live school. Live-school scholars take this same course on Optima's VR campus with a teacher and classmates, meeting daily as a core course.

Source: Algebra 1 Scope and Sequence (Optima Academy Online) · Florida course 1200310 / Honors 1200320 · Time commitment from the Optima master schedule: 250 class minutes weekly · Standards: Florida B.E.S.T. (MA.912).